

NAME $\Rightarrow$ TAHA FAYYAZ

S-62

XI

STATS NOTES

ROLL NO 27 (200702)

STATISTICS-XI

$\rightarrow$ measurement of scale

STATISTICS

FEDERAL BAORD

F.A./F. Sc./I.C.S. (First Year)

nominal
ordinal
interval
ratio

True
zero point
Temp

True
zero point
height
weight

INTRODUCTION

Origin of the word Statistics:

The word "Statistics" is comes from:

- The Latin word "Status" which means a political state or information useful for state.
- The Italian word "Statista" which means a political state or information useful for state.
- The German word "Statistik" which means a political science of several states.
- The French word "Statistique" which means a political state.

Meanings of the word Statistics:

- In plural form: Statistics is the aggregate of facts (data) in any field of life for specific purpose.
- In singular form: (As a Subject). Statistics is the science of systematic collection, presentation, analysis and interpretation of data to draw conclusions and to make decisions regarding population on the basis of sample data.
- As the plural of the word statistic. Statistic is a sample measure used to estimate the characteristics of the population. Statistics is the plural of the word statistic.

Characteristics of Statistics(i.e. Data):

- Statistics in the plural sense i.e. data has the following characteristics.
- Statistics are the aggregate of facts.
- Statistics are greatly affected by multiplicity of causes.
- Statistics are numerically expressed.
- Statistics are enumerated and estimated according to reasonable standards of accuracy.
- Statistics are collected in a systematic manner.
- Statistics are collected to fulfil a definite purpose.
- Statistics must be comparable to each other.

Functions/Uses of Statistics(i.e. as a Subject):

- Statistics in the singular sense i.e. as a subject has the following functions or uses.
- Statistics simplifies the complexities of huge mass of the data.
- Statistics present the facts in a more understandable numerical form.
- Statistics simplifies the comparison of the data.
- Statistics studies the relationship among the different facts.
- Statistics studies the changes in a variable or a group of related variables.
- Statistics helps us in forecasting.
- Statistics guides the managers in the formulation of policies.
- Statistics tests the laws of other sciences.

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Limitations of statistics:

- Statistics deals with the aggregates of facts, it does not deal with a single observation.
- Statistical results are valid on the average or in the long run.
- Statistics deals with the numerical or quantitative data.
- Statistical results are fallacious and misleading if sufficient care is not exercised in the collection, analysis and interpretation of the data.
- A person having expert knowledge of statistics is required to handle the data efficiently.
- Statistics is used only for analysis of data. It cannot change the nature of causes affecting the statistical data.

Population:

The whole aggregate of material possessing some common characteristics in any field of inquiry is called population.

Sample: |

A small representative part of the population is called sample.

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Parameter:

Any numerical value (i.e. Mean, Median, Mode, Variance, Standard Deviation etc.), calculated from the population data is called parameter, e.g. Mean (μ), Variance (σ^2) etc.

Statistic:

Any numerical value (i.e. Mean, Median, Mode, Variance, Standard Deviation etc.), calculated from the sample data is called statistic, e.g. Mean ($\bar{X}$) and Variance (S^2) etc.

Descriptive Statistics:

The branch of statistics which deals with collection, presentation and analysis of data is called descriptive statistics. Avg score

Inferential Statistics:

The branch of statistics which deals with making inferences, drawing conclusion and making decisions about any population parameter on the basis of sample data is called inferential statistics.

Variable: |

Any characteristics (quantitative/qualitative) which changes its value with respect to time, place or individual is known as variable e.g. marks obtained by the student, temperature of a place, height of a person, salary of Govt. Servants. Variables are denoted by capital letters say X, Y and Z.

Constant:

Any value, which does not change but remains fixed, is called a constant e.g. $\pi = 3.14159$ and $e = 2.71828$

Types of Data/Variable by Nature:

1-Qualitative Data/Variable:

A variable which changes only in quality (incapable of numerical measurement) from one individual to another individual is called a qualitative variable, for example marital status (married, single, widow, divorced) and colour of hair etc.

2-Quantitative Data/Variable:

A variable which changes only in quantity from one individual to another individual is called a quantitative variable e.g. Heights, Weights and Ages of different individuals, per capita income of different countries, imports and exports of different countries, production of different companies. The quantitative variable can be further divided into discrete and continuous variables.

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i-Discrete Data/Variable:

A quantitative variable which takes countable or finite number of values is called discrete variable. Such variables have a certain gap between two values and usually generated by counting process e.g. Number of students in different classes of a college, Number of children in different families, Number of vehicles registered in different districts of Pakistan etc.

ii-Continuous Data/Variable:

A quantitative variable which can assume/take each and every possible value between the two values is known as continuous variable. Such variables are generated by measuring process. e.g., Heights, Weights and Ages of different students and temperature of a place at different timings.

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Types of Data by Source:**Primary Data:**

The data originally collected from the prime source by some individual or agency is called primary data.

Secondary Data:

The data which is not originally collected from the prime source is called secondary data. Usually the data collected from the paper, internet, magazines, journals or other agency is called secondary data.

Types of Data by Treatment:**Raw Data:**

The data which is original in character, (Neither arranged nor classified in groups) and have not undergone any statistical treatment/tool is called raw data or ungrouped data.

Array Data:

When the raw data is arranged in ascending or descending order but not classified in groups, is called array data.

Grouped Data:

When the raw data is classified (divided in the groups) as per their resemblances and differences in a precise and more understandable way it is called grouped data.

Methods of collecting the Primary data: 2

- i. Direct personal observation.
- ii. Indirect oral investigation.
- iii. Registration.
- iv. Through enumerators.
- v. Estimates through local correspondents.

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Sources of collecting the Secondary data:

- i. **Official:** Federal bureau of statistics, population census organization, ministries of finance, commerce, agriculture, education and health etc.
- ii. **Semi-official:** State Bank of Pakistan, PEPCO, PTCL, IRSA etc
- iii. **Private:** Chambers of commerce and industry, other trade organizations and market committees.
- iv. Through the publications of research organizations i.e. PIDE, PARC, and different universities like Quaid-i-Azam university, Punjab university etc.
- v. Through magazines, journals and news papers.
- vi. Through inter-net by using different web-sites.

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Manifold Classification:

If the data is classified on the basis of more than two attributes it is called manifold classification.

Tabulation:

A systematic arrangement of the data into columns and rows is called tabulation. There may be simple, double or complex tabulation.

Main parts of statistical table:

Title: Clear-cut heading of table is called title.

Head Note/Prefatory Note: If title desires some further description, it should be mentioned after the title and is called head note or prefatory note.

Box Head: Headings and sub-headings in columns are called captions. Spaces where captions are written are called box-head.

Stub: Headings and sub-headings in rows and row captions. Space where row-captions are written is called stub.

Body: The numerical data below the column-captions and besides the row-captions are body of the table.

Foot Notes: Some notes are sometimes necessary for explanation. If they are under the title they are prefatory notes and if they are at the end they are foot notes.

Source Notes: Source notes are at the end of the table. They generally give source of data.

Frequency Distribution:

Frequency distribution is a tabular arrangement of data in which various items are arranged into classes (prepared on the basis of numerical values) and the number of items falling in each class.

Main steps involved in the construction of a frequency distribution:

- Identify the smallest and largest values of the data and find out the range by subtracting the smallest from the largest one.
- Decide on the number of classes:
$$\text{No. of classes} = 1 + 3.3 \log(n)$$
 Where 'n' is the number of values in the data.
- Determine the class interval by dividing the range by the number of classes.
- Decide about the lower class limit/boundary of the lowest class.
- Find the upper class boundary of the lowest class by adding class interval to the lower class boundary, and then determine the other classes by adding the class interval.
- Distribute the individual observations into classes either by listing the values or by using tally marks.
- Then finally count the values falling in each class and write the figure in front of the respective classes. These are the frequencies of each class.

Class Limits:

If both the upper and lower values describing a class are included in that particular class they are called class limits e.g. 31-40, 41-50, 51-60 etc. This is called inclusive style of making the groups.

Class Boundaries:

The true or precise values, which describe the actual or true class limits of a class, are called class boundaries e.g. 30-40, 40-50, 50-60 or 20.5-30.5, 30.5-40.5, 40.5-50.5 etc. In this case the upper class boundary is not included in that particular class and this is called exclusive style of making the groups.

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Mid Point/Class Mark:

It is the point, which divides the class into two equal halves. It is calculated by finding the mean/average of the two class limits or two class boundaries of the same class.

Class Frequency:

The number of items falling in a particular class is called the frequency of that class.

Class Interval:

The difference between the upper and lower class boundaries (Not class limits) of a class is called class interval.

Relative Frequency:

If frequency of a class is divided by the sum of all frequencies then it is called relative frequency or proportion or probability of that class. Total of relative frequency is always 1.

Percentage Relative Frequency:

If the relative frequency is multiplied by 100 we get percentage relative frequency.

Cumulative Frequency Distribution OR

Less Than Cumulative Frequency Distribution:

Total number of values less than the upper class boundary of a particular class is called cumulative frequency or less than cumulative frequency of that particular class. The table showing the cumulative frequencies against the class boundaries of each class is called cumulative frequency distribution or less than cumulative frequency distribution.

More Than Cumulative Frequency Distribution:

Total number of values greater than the lower class boundary of a particular class is called more than cumulative frequency of that particular class. The table showing more than cumulative frequencies against the class boundaries of each class is called more than cumulative frequency distribution.

Bi-variate Frequency Distribution:

Sometimes we have observations in pairs and each pair is obtained as a result of observations on two variables. Tabular arrangement of bi-variate data into cross classification is known as bi-variate or two-way frequency distribution.

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DIAGRAMS

Simple Bar Diagram/Chart:

A simple bar chart/diagram is used to represent the data involving only one variable classified on spatial, qualitative or temporal basis. In these diagrams we make bars of equal width but variable length.

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Multiple Bar Diagrams/Charts:

Multiple bar diagrams are used when the data in the form of different groups and each group contains two or more pieces of information and their total is not required. Within the group, the different quantities are represented by adjacent bars and different groups are apart from each other.

Component (Sub-Divided) Bar Diagram/Chart:

When it is desired to compare the totals of some groups of data and the comparison within the group is also required then component/sub-divided bar diagram or chart is used. Within the group, the different quantities are represented by dividing the bar in parts, proportional to the number of items in each subgroup.

Percentage Component Bar Chart:

Component bar charts may also be drawn on percentage basis by expressing the components as a percentage of their respective totals. It is called percentage component bar chart.

Rectangular Diagrams:

Rectangular diagrams are drawn for those quantities which differ a lot from one another. These are the area diagrams. Data is represented by the area of rectangle (Length $\times$ width). One of the two dimensions of rectangle i.e. either length or breadth kept constant and the other is taken proportional to the size of the data.

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Sub-Divided Rectangles:

The sub-divided rectangles are drawn for presentation of two or more groups of data when each group consists of some components and these components are also to be compared. These are usually drawn to compare the budgets of various families.

Pie Diagram/Chart:

Pie diagram is a circular diagram where the whole circle represents a "total" and the components of the total are represented by sectors of the circle. It is also called a circular diagram. As a circle consists of 360° of angle therefore to show the components by sectors, we calculate the angles for each sector by the formula:

$$\text{Angle of the Sector} = \frac{[\text{Component part} \times 360]}{\text{Total}}$$

If we have to compare the groups by pie chart then the area of the circle is proportional to the total of the group.

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GRAPHS

Historigram:

The graphical representation of a time series is called "Historigram". The time is taken along X-axis and the variable is taken along Y-axis. The plotted points are joined together to get a line graph, which is called "Historigram".

Graphs of Frequency Distribution:

Histogram:

A histogram consists of a set of adjacent rectangles having bases along the X-axis (Marked off by class boundaries) and areas proportional to the class frequencies. i.e.

Area $\propto$ Frequency

(Height of the bar $\times$ Class width) = Frequency

Height of the bar = (Frequency / Class width) = (f/h)

To draw a histogram (for equal class intervals) class boundaries (not class limits) are marked along the X-axis and rectangles are constructed with widths equal to class intervals and heights proportional to class frequencies. To adjust the heights of the rectangles in frequency distributions with unequal class intervals, each class frequency is divided by its class interval.

Frequency Polygon:

It is a line graph constructed by plotting the class frequencies against their corresponding class marks (mid points) and then joining the resulting points by means of straight lines.

Relative Frequency Histogram:

A relative frequency histogram consists of a set of adjacent rectangles having base along the X-axis (Marked off by class boundaries) and area proportional to the class relative frequencies.

Relative Frequency Polygon:

It is a line graph constructed by plotting the class relative frequencies against their corresponding class marks (Mid-points) and then joining the resulting points by straight lines.

Frequency Curve:

If the lines of a frequency polygon are smoothed so that there are no sharp ups and downs so as results into a continuous smooth curve then it is called a frequency curve.

Cumulative Frequency Polygon or Ogive or Less Than Cumulative Frequency Polygon or Less Than Ogive:

A line graph representing the less than cumulative frequencies plotted against the upper class boundaries is called a cumulative frequency polygon or ogive.

More Than Cumulative Frequency Polygon or More Than Ogive:

A line graph representing the more than cumulative frequencies plotted against the lower class boundaries is called a more than cumulative frequency polygon or more than ogive.

Cumulative Frequency Curve:

If the points plotted for a cumulative frequency polygon are smoothed out, the resulting curve is called cumulative frequency curve.

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MEASURES OF CENTRAL TENDENCY

Measure of Central Tendency or Average:

The numerical measure which is used to represent the entire data is called average. The averages usually lie in the centre of the data that is why they are called measures of central tendency or measures of location. Most common of these measures are Arithmetic Mean, Median, Mode, Geometric mean and Harmonic mean.

OR The numerical measure which represents the centre of the data is called measure of central tendency or average.

Properties/Qualities of a good average:

1. It should be easy to understand and simple to calculate.
2. It should be clearly defined by a mathematical formula.
3. It should be based on all the observations of the data.
4. It should not be affected by extreme values.
5. It should be capable for further mathematical treatment.
6. It should give stable results in repeated sampling experiments.

Arithmetic Mean or Mean:

The arithmetic mean or mean is defined as a value obtained by dividing the sum of the values by the number of values.

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} = \frac{\sum x}{n} \quad (\text{For ungrouped data})$$

$$\bar{X} = \frac{\sum (fx)}{\sum f} \quad (\text{For discrete or continuous grouped data})$$

Weighted Arithmetic Mean:

If we want to find out the arithmetic mean of values, which are not of equal importance, then we assign them certain numerical values to express their relative importance these numerical values are called weights. If $x_1, x_2, x_3, \dots, x_k$ have weights $w_1, w_2, w_3, \dots, w_k$ then the weighted arithmetic mean $\bar{X}_w$ is defined as:

$$\bar{X}_w = \frac{w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_k x_k}{w_1 + w_2 + w_3 + \dots + w_k} = \frac{\sum w_j x_j}{\sum w_j} = \frac{\sum wx}{\sum w}$$

Combined Mean:

If $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$ be the arithmetic means of k distributions with respective frequencies $n_1, n_2, n_3, \dots, n_k$ the combined arithmetic mean $\bar{X}_c$ is defined as:

$$\bar{X}_c = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + n_3 \bar{X}_3 + \dots + n_k \bar{X}_k}{n_1 + n_2 + n_3 + \dots + n_k} = \frac{\sum n_j \bar{X}_j}{\sum n_j} = \frac{\sum n_j \bar{X}_j}{\sum n_j}$$

Effect of change of origin and scale on arithmetic mean:

Change of origin: To simplify our calculations sometimes we subtract (or add) a constant from all the values of the variable X i.e. $D = (X - A)$. Then the arithmetic mean of the new variable D would also be decreased by that constant as compare to the arithmetic mean of the original variable X . So, The arithmetic mean of X can be calculated by the following formulae:

$$\text{As } D = X - A$$

$$\text{So } X = A + D$$

$$\text{Hence } \bar{X} = A + \bar{D} = A + \left(\frac{\sum d}{n} \right) \quad (\text{For ungrouped data})$$

$$\text{And } \bar{X} = A + \bar{D} = A + \left(\frac{\sum fd}{\sum f} \right) \quad (\text{For grouped data.})$$

This method is called indirect method or shortcut method.

Change of origin and scale: Sometimes, to simply our calculations, it is not enough to subtract a particular value from the data and we also have to divide each value of the variable by a suitable constant. In result of subtracting a constant from every value of the variable and then dividing by a constant a new variable U has been introduced in such a way that:

$$U = \left(\frac{X - A}{h} \right) \quad \text{or} \quad X = A + hU$$

We can find out the arithmetic mean of X , using the variable U by the following formulae:

$$\bar{X} = A + h\bar{U} = A + h \left(\frac{\sum u}{n} \right) \quad (\text{For ungrouped data})$$

$$\bar{X} = A + h\bar{U} = A + h \left(\frac{\sum fu}{\sum f} \right) \quad (\text{For grouped data.})$$

This method is also called coding method or step deviation method.

Properties of Arithmetic Mean: *

1. The mean of the constant is that constant itself, i.e. $\bar{K} = K$
2. Sum of deviation from arithmetic mean is always zero, i.e. $\sum (x - \bar{X}) = 0$
3. Sum of squares of deviations from arithmetic mean is minimum (least).
i.e. $\sum (x - \bar{X})^2 \leq \sum (x - A)^2$

It is called the minimal property of arithmetic mean.

4. The arithmetic mean is affected by change of origin and scale. Let $x_1, x_2, x_3, \dots, x_n$ be a sample of n values having mean $\bar{X}$. If $Y = a + bX$, then $\bar{Y} = a + b\bar{X}$.
5. If $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$ be the arithmetic means of k groups with respective frequencies $n_1, n_2, n_3, \dots, n_k$ the combined arithmetic mean $\bar{X}$ is defined as:

$$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + n_3\bar{X}_3 + \dots + n_k\bar{X}_k}{n_1 + n_2 + n_3 + \dots + n_k} = \frac{\sum_{j=1}^k n_j \bar{X}_j}{\sum_{j=1}^k n_j} = \frac{\sum n_j \bar{X}_j}{\sum n_j}$$

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Merits of Arithmetic Mean:

1. It is easy to calculate and simple to understand.
2. It is clearly defined by mathematical formula.
3. It is based on all the observations of the variable.
4. It is capable for further mathematical treatment and analysis.
5. It provides a good basis for comparison.

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Demerits of Arithmetic Mean:

1. It is highly affected by extreme values.
2. It cannot be calculated for qualitative data.
3. It cannot be calculated for open ended frequency distribution.
4. It is not a suitable average for ratios and percentages.

Geometric Mean:

Geometric mean is the n^{th} positive root of the product of 'n' positive values.

$$G.M = \sqrt[n]{(x_1)(x_2)(x_3) \dots (x_n)}$$

(for ungrouped data)

$$G.M = [(x_1)(x_2)(x_3) \dots (x_n)]^{1/n}$$

(for grouped data)

$$G.M = [(x_1)^{f_1} \cdot (x_2)^{f_2} \cdot (x_3)^{f_3} \dots (x_k)^{f_k}]^{1/(\sum f)}$$

(for ungrouped data)

OR. $G.M = \text{Anti log} \left[\frac{\sum \log(x)}{n} \right]$

(for grouped data)

$$G.M = \text{Anti log} \left[\frac{\sum [f \log(x)]}{\sum f} \right]$$

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Merits of Geometric Mean:

1. It is clearly defined by a mathematical formula.
2. It is based on all the observations of the variable.
3. It is capable for further mathematical treatment.
4. It gives importance to all rates of change that is why it is less affected by extreme values.
5. It is not much affected by the fluctuations of sampling.
6. It is the most suitable average for ratios, percentages and rates of increase or decrease.

Demerits of Geometric Mean:

1. It is neither simple to understand nor easy to calculate.
2. It cannot be calculated if any of the values in the data is zero or negative.

Harmonic Mean:

Harmonic mean is the reciprocal of the arithmetic mean of the reciprocals of non-zero (preferably positive) values.

$$H.M = \frac{n}{\sum (1/x)}$$

(for ungrouped data)

$$H.M = \frac{\sum f}{\sum (f/x)}$$

(for grouped data)

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Merits of Harmonic Mean:

1. It is clearly defined by a mathematical formula.
2. It is based on all the observations of the variable.
3. It is stable and not much affected by the fluctuations of sampling.
4. It is the most suitable average for rates (speed/hr etc.) and ratios.

Demerits of Harmonic Mean:

1. It is neither simple to understand nor easy to calculate.
2. It cannot be calculated if any of the values in the data is zero.
3. It gives too much importance to small values that is why it is used only in special cases.

Note: There is very interesting relationship between the Arithmetic mean, Geometric mean and Harmonic mean of the same set of data which is:

$$\text{A.M} \geq \text{G.M} \geq \text{H.M} \\ \text{OR} \quad \text{H.M} \leq \text{G.M} \leq \text{A.M}$$

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Median:

Median is the middle most/central most value of the observations arranged in ascending or descending order. OR Median is the exact central value of an array data/distribution.

$$\text{Median} = \tilde{X} = \text{Value of } \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item} \quad (\text{for ungrouped data})$$

$$\text{Median} = \tilde{X} = l + \frac{h}{f} \left(\frac{n}{2} - C \right) \quad (\text{for grouped data})$$

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Merits of Median:

1. It is easy to calculate and simple to understand.
2. It is not affected by extreme values.
3. Being a positional average it can be calculated for open ended frequency distribution.
4. It is suitable average for qualitative data.
5. It can be located graphically.
6. It is a suitable average for highly skewed distributions.

Demerits of Median:

1. Array data is required for median which is difficult and time consuming task.
2. It is not capable for further mathematical treatment.
3. It is not based on all the observations and ignores the extreme values.

Mode:

Mode is the most frequent observation of the data if such an observation exists. For discrete data mode can be found just by inspection. Mode is the value with maximum frequency.

$$\text{Mode} = \hat{X} = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h \quad (\text{for grouped data})$$

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Merits of Mode:

1. It is easy to calculate and simple to understand. It can be found just by inspection of ungrouped data and discrete frequency distribution.
2. It is not affected by extreme values.
3. It can be calculated for open ended frequency distribution.
4. It is suitable average for qualitative data.
5. It can be located graphically.

Demerits of Mode:

1. It is not based on all the observations.
2. It is not capable for further mathematical treatment.
3. It is stable average for large data, but may not be stable for small data.
4. Some times mode does not exist in the data, and sometimes it has more than one value, so it is difficult to find the true value of mode in that kind of situations.

Note: Mode is the only measure of central tendency which can have more than one value.

How to decide a suitable average?

All the measures of central tendency are different in nature and have their own specific properties and method of calculation. That is the reason; we cannot use every measure of central tendency in all the situations. The following are some typical situations in which any measure of central tendency is used appropriately.

- Arithmetic Mean** is a suitable average for the quantitative data which is homogeneous in nature; means there are no extreme values in the data. When the rates are expressed as x per y and y is constant, then arithmetic mean is suitable.
- Median** is a suitable average for the qualitative data. It is also a suitable average for the heterogeneous quantitative data which means there are some extreme values present in the data. It is suitable for skewed distributions as well.
- Mode** is a suitable for qualitative data. It is considered as the best average for the quantitative data where people likes and dislikes is involved.
- Geometric Mean** is appropriate to average the ratios and rates of change, i.e. rate of increase or rate of decrease etc.
- Harmonic Mean** is suitable to average the different speeds in different intervals of equal distance. When the rates are expressed as x per y and x is constant, the harmonic mean is suitable.

Quartiles:

The three values which divide the data, arranged in ascending order, into four equal parts, in such a way that each part contains equal number of values, are known as quartiles.

$$Q_1 = \text{Value of } \left(\frac{n+1}{4} \right)^{\text{th}} \text{ item}$$

$$Q_2 = \text{Value of } \left(\frac{2(n+1)}{4} \right)^{\text{th}} \text{ item}$$

$$Q_3 = \text{Value of } \left(\frac{3(n+1)}{4} \right)^{\text{th}} \text{ item}$$

(for ungrouped data)

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$$Q_1 = l + \frac{h}{f} \left(\frac{n}{4} - C \right)$$

$$Q_2 = l + \frac{h}{f} \left(\frac{2n}{4} - C \right)$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3n}{4} - C \right)$$

(for grouped data)

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Deciles:

The deciles are the nine values, which divide the data, arranged in ascending order, in ten equal parts in such a way that each part contains equal number of values. They are denoted by $D_1, D_2, \dots, D_9$.

$$D_k = \text{Value of } \left(\frac{k(n+1)}{10} \right)^{\text{th}} \text{ item} \quad \text{Where } k = 1, 2, 3, \dots, 9 \quad (\text{for ungrouped data})$$

$$D_k = l + \frac{h}{f} \left(\frac{kn}{10} - C \right) \quad \text{Where } k = 1, 2, 3, \dots, 9 \quad (\text{for grouped data})$$

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Percentiles:

The ninety-nine (99) values, which divide the data, arranged in ascending order, in hundred equal parts in such a way that each part contains equal number of values, are called percentiles. They are denoted by $P_1, P_2, \dots, P_{99}$.

$$P_k = \text{Value of } \left(\frac{k(n+1)}{100} \right)^{\text{th}} \text{ item} \quad \text{Where } k = 1, 2, 3, \dots, 99 \quad (\text{for ungrouped data})$$

$$P_k = l + \frac{h}{f} \left(\frac{kn}{100} - C \right) \quad \text{Where } k = 1, 2, 3, \dots, 99 \quad (\text{for grouped data})$$

Note:

$$Q_1 = P_{25}$$

$$Q_2 = \text{Median} = P_{50} = D_5$$

$$Q_3 = P_{75}$$

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Empirical relation between Mean, Median and Mode:

There exists a very important relationship between Mean, Median and Mode calculated from a data set.

- If the distribution is symmetrical mean, median and mode coincide/(are equal): i.e. they have same value.

$$\text{Mean} = \text{Median} = \text{Mode}$$

- If the distribution is moderately skewed then:

$$(\text{Mean} - \text{Mode}) = 3(\text{Mean} - \text{Median})$$

$$\text{Mean} - \text{Mode} = 3\text{Mean} - 3\text{Median}$$

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

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MEASURES OF DISPERSION

Dispersion:

By dispersion we mean the extent to which the values are spread out from the average. The measures used for this purpose are called measures of dispersion or measures of variation.

Absolute Measure of Dispersion:

The actual dispersion or variation, which is measured in the same units as of the original data, is called absolute dispersion. Absolute measures of dispersion cannot be used to compare two sets of data of different units or even if the two sets of the data in the same units.

Relative Measure of Dispersion:

The dispersion, which is expressed in the form of the ratio, co-efficient or percentage and is free from any unit of measurement, is called relative dispersion. The measures which measure this kind of dispersion are called relative measure of dispersion. These measures are usually used for comparison of variation in different kind of data.

Absolute Measures Of Dispersion:

Range:

Range is the difference of largest and smallest value of the data or distribution.

$$R = x_m - x_0$$

Quartile Deviation Or Semi Inter Quartile Range:

It is defined as the half of the difference between the upper and lower quartiles.

$$Q.D = S.I.Q.R = \frac{Q_3 - Q_1}{2}$$

Mean Deviation:

Mean deviation is the arithmetic mean of the absolute deviations from some central value say mean, median, or mode. If $\bar{X}$ = Mean, $\tilde{X}$ = Median and $\hat{X}$ = Mode, then

$$M.D = \frac{\sum |x - \bar{X}|}{n}$$

$$M.D = \frac{\sum |x - \tilde{X}|}{n}$$

$$M.D = \frac{\sum |x - \hat{X}|}{n}$$

$$M.D = \frac{\sum f|x - \bar{X}|}{\sum f}$$

$$M.D = \frac{\sum f|x - \tilde{X}|}{\sum f}$$

$$M.D = \frac{\sum f|x - \hat{X}|}{\sum f}$$

(for ungrouped data)

(for grouped data)

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Variance:

Variance is defined as the arithmetic mean of the squared deviations of the values taken from their arithmetic mean. Population variance is denoted by σ^2 and sample variance by S^2 .

$$S^2 = \frac{\sum (x - \bar{X})^2}{n} = \left[\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \right] = \left[\frac{\sum x^2}{n} - (\bar{X})^2 \right] \quad (\text{for ungrouped data})$$

$$S^2 = \frac{\sum f(x - \bar{X})^2}{\sum f} = \left[\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 \right] = \left[\frac{\sum fx^2}{\sum f} - (\bar{X})^2 \right] \quad (\text{for ungrouped data})$$

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Standard Deviation

The positive square root of the arithmetic mean of squared deviations of the values taken from their arithmetic mean is called standard deviation. It is actually positive square root of the variance. Population standard deviation is denoted by σ and sample standard deviation by S .

$$S = \sqrt{\frac{\sum (x - \bar{X})^2}{n}} = \sqrt{\left[\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \right]} = \sqrt{\left[\frac{\sum x^2}{n} - (\bar{X})^2 \right]} \quad (\text{for ungrouped data})$$

$$S = \sqrt{\frac{\sum f(x - \bar{X})^2}{\sum f}} = \sqrt{\left[\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 \right]} = \sqrt{\left[\frac{\sum fx^2}{\sum f} - (\bar{X})^2 \right]} \quad (\text{for grouped data})$$

Affect of change of origin and scale on variance and standard deviation:

Change of origin The value of variance and standard deviation does not change if a constant value is added or subtracted from the data.

Let $D = X - A$

or $X = A + D$

$$S_x^2 = S_D^2 = \frac{\sum (d - \bar{D})^2}{n} = \left[\frac{\sum d^2}{n} - \left(\frac{\sum d}{n} \right)^2 \right] \quad (\text{for ungrouped data})$$

$$S_x^2 = S_D^2 = \frac{\sum f(d - \bar{D})^2}{\sum f} = \left[\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f} \right)^2 \right] \quad (\text{for grouped data})$$

Similarly the standard deviation is:

$$S_x = S_D = \sqrt{\frac{\sum (d - \bar{D})^2}{n}} = \sqrt{\left[\frac{\sum d^2}{n} - \left(\frac{\sum d}{n} \right)^2 \right]} \quad (\text{for ungrouped data})$$

$$S_x = S_D = \sqrt{\frac{\sum f(d - \bar{D})^2}{\sum f}} = \sqrt{\left[\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f} \right)^2 \right]} \quad (\text{for grouped data})$$

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Change of scale: The values of variance and standard deviation are affected by change of scale. OR If all the values of the variable are multiplied or divided by a constant then variance or standard deviation of the new variable is changed.

Let $Y = \frac{X}{b}$ or $X = bY$

Then

$$S_y^2 = b^2 S_x^2 = b^2 \left[\frac{\sum (y - \bar{Y})^2}{n} \right] = b^2 \left[\frac{\sum y^2}{n} - \left(\frac{\sum y}{n} \right)^2 \right]$$

(for ungrouped data)

$$S_y^2 = b^2 S_x^2 = b^2 \left[\frac{\sum f(y - \bar{Y})^2}{\sum f} \right] = b^2 \left[\frac{\sum fy^2}{\sum f} - \left(\frac{\sum fy}{\sum f} \right)^2 \right]$$

(for grouped data)

Similarly the standard deviation is:

$$S_y = |b| S_x = |b| \sqrt{\left[\frac{\sum (y - \bar{Y})^2}{n} \right]} = |b| \sqrt{\left[\frac{\sum y^2}{n} - \left(\frac{\sum y}{n} \right)^2 \right]}$$

(for ungrouped data)

$$S_y = |b| S_x = |b| \sqrt{\left[\frac{\sum f(y - \bar{Y})^2}{\sum f} \right]} = |b| \sqrt{\left[\frac{\sum fy^2}{\sum f} - \left(\frac{\sum fy}{\sum f} \right)^2 \right]}$$

(for grouped data)

Change of origin and scale: If the change of origin and scale are applied simultaneously then: Let $U = \frac{X - A}{h}$ or $X = A + hU$

Hence

$$S_u^2 = h^2 S_x^2 = h^2 \left[\frac{\sum (u - \bar{U})^2}{n} \right] = h^2 \left[\frac{\sum u^2}{n} - \left(\frac{\sum u}{n} \right)^2 \right]$$

(for ungrouped data)

$$S_u^2 = h^2 S_x^2 = h^2 \left[\frac{\sum f(u - \bar{U})^2}{\sum f} \right] = h^2 \left[\frac{\sum fu^2}{\sum f} - \left(\frac{\sum fu}{\sum f} \right)^2 \right]$$

(for grouped data)

"h" is usually common class interval which is positive so the standard deviation is:

$$S_u = h S_x = h \sqrt{\left[\frac{\sum (u - \bar{U})^2}{n} \right]} = h \sqrt{\left[\frac{\sum u^2}{n} - \left(\frac{\sum u}{n} \right)^2 \right]}$$

(for ungrouped data)

$$S_u = h S_x = h \sqrt{\left[\frac{\sum f(u - \bar{U})^2}{\sum f} \right]} = h \sqrt{\left[\frac{\sum fu^2}{\sum f} - \left(\frac{\sum fu}{\sum f} \right)^2 \right]}$$

(for grouped data)

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Relative Measures of Dispersion:

Co-efficient of Range:

It is the relative measure of dispersion of range and defined as:

$$\text{Co-efficient of Range} = \frac{(x_m - x_o)}{(x_m + x_o)}$$

Co-efficient of Quartile Deviation:

It is the relative measure of quartile deviation and defined as:

$$\text{Co-efficient of Q.D.} = \frac{(Q_3 - Q_1)}{(Q_3 + Q_1)}$$

Co-efficient of Mean Deviation:

Co-efficient of mean deviation is defined as a ratio of mean deviation and used measure of central tendency.

$$\text{Co-efficient of M.D.} = (\text{M.D from Mean}) / \text{Mean}$$

$$\text{Co-efficient of M.D.} = (\text{M.D from Median}) / \text{Median}$$

$$\text{Co-efficient of M.D.} = (\text{M.D from Mode}) / \text{Mode}$$

Co-efficient of Variation:

Co-efficient of variation is the relative measure of dispersion and defined as the percentage ratio of the standard deviation to the arithmetic mean.

$$C.V = \frac{S}{X} \times 100$$

Co-efficient of Standard Deviation:

Co-efficient of standard deviation is defined as the ratio of standard deviation to arithmetic mean. Co-efficient of S.D = $\frac{S}{X}$

Properties of Variance:

- $\text{Var}(C) = 0$ *constant*
- $\text{Var}(X \pm a) = \text{Var}(X)$
- $\text{Var}(ax) = a^2 \text{Var}(X)$
- If there are K groups with observations $n_1, n_2, n_3, \dots, n_k$ and means $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k$ and variances $S_1^2, S_2^2, \dots, S_k^2$ then the combined variance is

$$S_c^2 = \frac{\sum_{j=1}^k [n_j \{S_j^2 + (\bar{X}_j - \bar{X}_c)^2\}]}{\sum n_j}$$

- $\text{Var}(X \pm Y) = [\text{Var}(X) + \text{Var}(Y)]$
- Variance is a positive quantity and expressed in the square units of data.

$$S_c^2 = \frac{1}{\sum n_j} \sum_{j=1}^k [n_j \{S_j^2 + (\bar{X}_j - \bar{X}_c)^2\}]$$

Properties of Standard Deviation:

- $\text{SD}(C) = 0$
- $\text{SD}(X \pm a) = \text{SD}(X)$
- $\text{SD}(ax) = |a| \text{SD}(X)$
- If there are K groups with observations $n_1, n_2, n_3, \dots, n_k$ and means $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k$ and variances $S_1^2, S_2^2, \dots, S_k^2$ then the combined standard deviation is

$$S_c = \sqrt{\frac{\sum [n_j \{S_j^2 + (\bar{X}_j - \bar{X}_c)^2\}]}{\sum n_j}}$$

- $\text{SD}(X \pm Y) = \sqrt{[\text{Var}(X) + \text{Var}(Y)]}$
- S.D is a positive quantity and expressed in the same units as data.

$$S_c = \frac{1}{\sum n_j} \sum_{j=1}^k [n_j \{S_j^2 + (\bar{X}_j - \bar{X}_c)^2\}]$$

MOMENTS

The r^{th} moment is the arithmetic mean of the r^{th} power of deviations (from mean or arbitrary mean or origin i.e. zero). We now discuss different types of moments separately.

Moments about origin:

The r^{th} moment about origin is the arithmetic mean of r^{th} power of the variable. These moments are denoted by μ'_r for population data and m'_r for sample data. Mathematically:

$$m'_r = \frac{\sum x^r}{n} \quad (\text{For ungrouped data})$$

$$m'_r = \frac{\sum fx^r}{\sum f} \quad (\text{For grouped data})$$

The first four moments about origin for sample data are:

For ungrouped data	For grouped data
$m'_1 = \frac{\sum x}{n}$	$m'_1 = \frac{\sum fx}{\sum f}$
$m'_2 = \frac{\sum x^2}{n}$	$m'_2 = \frac{\sum fx^2}{\sum f}$
$m'_3 = \frac{\sum x^3}{n}$	$m'_3 = \frac{\sum fx^3}{\sum f}$
$m'_4 = \frac{\sum x^4}{n}$	$m'_4 = \frac{\sum fx^4}{\sum f}$

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Moments about arbitrary origin:

The r^{th} moment about arbitrary origin (A) is the arithmetic mean of r^{th} power of the deviations taken from the arbitrary origin (provisional mean). These moments are denoted by μ'_r for population data and m'_r for sample data. Mathematically:

$$m'_r = \frac{\sum (x - A)^r}{n} \quad (\text{For ungrouped data})$$

$$m'_r = \frac{\sum f(x - A)^r}{\sum f} \quad (\text{For grouped data})$$

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The first four moments about arbitrary origin for the sample data are:

For ungrouped data	For grouped data
$m'_1 = \frac{\sum (x - A)}{n}$	$m'_1 = \frac{\sum f(x - A)}{\sum f}$
$m'_2 = \frac{\sum (x - A)^2}{n}$	$m'_2 = \frac{\sum f(x - A)^2}{\sum f}$
$m'_3 = \frac{\sum (x - A)^3}{n}$	$m'_3 = \frac{\sum f(x - A)^3}{\sum f}$
$m'_4 = \frac{\sum (x - A)^4}{n}$	$m'_4 = \frac{\sum f(x - A)^4}{\sum f}$

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Effect of change of origin and scale on moments about mean:

Change of origin: If m_r is the r^{th} mean moments of the variable X and a new variable introduced as $D = X - A$.

$$\text{Then } m_r = \frac{\sum (x - \bar{x})^r}{n}$$

$$\text{As } D = X - A$$

$$\text{So } X = A + D$$

$$\text{and } \bar{X} = A + \bar{D}$$

$$\text{hence } m_r = \frac{\sum [(A + D) - (A + \bar{D})]^r}{n}$$

$$m_r = \frac{\sum (D - \bar{D})^r}{n}$$

r^{th} moment of $X = r^{\text{th}}$ moment of D

Hence change of origin does not affect the central moments about mean.

Change of origin and scale: Let m_r is the r^{th} mean moments of the variable X . If we introduce a new variable to simplify our calculations as

$$U = \frac{X - A}{h} \quad \text{or} \quad X = A + hU$$

$$\text{Then } m_r = \frac{\sum (x - \bar{x})^r}{n}$$

$$\text{As } U = \frac{X - A}{h}$$

$$\text{so } X = A + hU$$

$$\text{and } \bar{X} = A + h\bar{U}$$

$$m_r = \frac{\sum [(A + hU) - (A + h\bar{U})]^r}{n}$$

$$m_r = h^r \left[\frac{\sum (u - \bar{u})^r}{n} \right]$$

$$m_r = h^r \cdot v_r \quad \text{Where } v_r = \left[\frac{\sum (u - \bar{u})^r}{n} \right]$$

r^{th} mean moment of $X = h^r$ (r^{th} mean moment of U)

Hence change of scale affects the moments about mean.

Calculation of Moments about mean using change of origin and scale:

Method-1 Let m_1, m_2, m_3 & m_4 are the mean moments of the variable X , and m'_1, m'_2, m'_3 & m'_4 are the moments about origin of the variable X .

First calculate m'_1, m'_2, m'_3 & m'_4 using the formulae:

$$m'_1 = h \times \left(\frac{\sum fu}{\sum f} \right), \quad m'_2 = h^2 \times \left(\frac{\sum fu^2}{\sum f} \right), \quad m'_3 = h^3 \times \left(\frac{\sum fu^3}{\sum f} \right), \quad m'_4 = h^4 \times \left(\frac{\sum fu^4}{\sum f} \right)$$

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$$\mu_3 = \frac{\sum (x - \bar{x})^3}{N}$$

$$b_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

Moment Ratios:

The most common moment ratios for population data are as follows:

$$\beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3} \quad \text{and} \quad \beta_2 = \frac{\mu_4}{(\mu_2)^2}$$

These moment ratios describe the skewness and kurtosis of the distribution which would be discussed later. Moment ratios for sample data are denoted by b_1 & b_2 and are as follows:

$$b_1 = \frac{(m_3)^2}{(m_2)^3} \quad \text{and} \quad b_2 = \frac{m_4}{(m_2)^2}$$

Roll of moments in describing a frequency distribution:

To describe a frequency distribution we need measure of central tendency, measure of dispersion, measure of skewness and measure of kurtosis. All of this information can be derived from the first four moments as:

1. The first moment about origin is arithmetic mean.
2. The second moment about mean is variance & its positive square root is standard deviation.
3. The third mean moment is a measure of skewness.
4. The fourth moment is used to measure kurtosis.

Thus the first four moments play a key roll in describing the frequency distributions.

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SKEWNESS:

If both the tails of a frequency curve are equal in length from the central point of the frequency curve then the distribution is said to be symmetrical. If any tail of the frequency curve is longer than the other one then the distribution is said to be asymmetrical or skewed. Thus skewness may be defined as departure from symmetry.

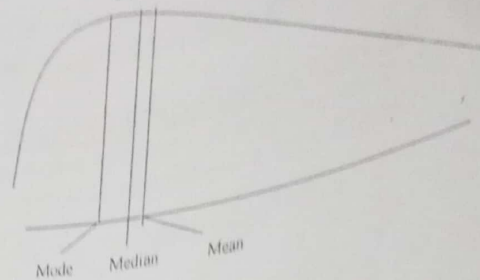
OR

ترتیب کی
Lack of symmetry is called skewness.

Positive Skewness:

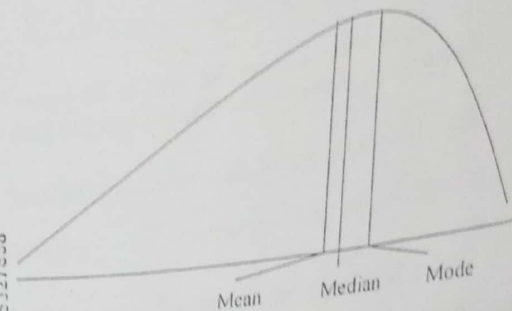
- A distribution is said to be positively skewed if:
- Mean > Median > Mode.
 - $(Q_3 - \text{Median}) > (\text{Median} - Q_1)$
 $(Q_1 + Q_3) > 2(\text{Median})$
 - $\Sigma(x - \text{Median}) > 0$
 - $m_3 > 0$
 - The right tail of the frequency curve is longer than the left tail of the curve.

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**Negative Skewness:**

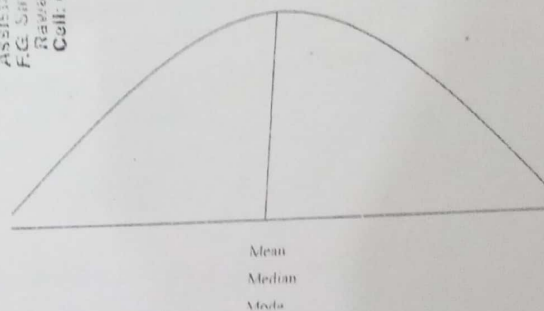
- A distribution is said to be negatively skewed if:
- Mean < Median < Mode
 - $(Q_3 - \text{Median}) < (\text{Median} - Q_1)$
 $(Q_1 + Q_3) < 2(\text{Median})$
 - $\Sigma(x - \text{Median}) < 0$
 - $m_3 < 0$
 - The left tail of the frequency curve is longer than the right tail of the curve.

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**Symmetrical Distribution:**

A distribution is said to be symmetrical if:

- Mean = Median = Mode
- $(Q_3 - \text{Median}) = (\text{Median} - Q_1)$
 $(Q_1 + Q_3) = 2(\text{Median})$
 $(\text{Median}) = (Q_1 + Q_3)/2$
- $\Sigma(x - \text{Median}) = 0$
- $m_3 = 0$
- Both the tails of frequency curve are equal in length.



Note: The first moment ratio is used to measure the skewness.

$$\beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3} \quad \text{and} \quad b_1 = \frac{(m_3)^2}{(m_2)^3}$$

If β_1 or $b_1 = 0$ Then distribution is symmetrical.

If β_1 or $b_1 \neq 0$ Then distribution is skewed but it does not tell us about positive or negative skewness.

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Karl Pearson's Co-efficient of Skewness:

The difference between mean and mode is an absolute measure of skewness. If we divide this difference by standard deviation, it is called a relative measure of skewness or co-efficient of skewness. This measure of skewness is suggested by Karl Pearson (1857-1936).

$$\text{Co-efficient of skewness} = \frac{(\text{Mean} - \text{Mode})}{S}$$

(Mean - Mode)

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If mode is not clearly located then:

$$\text{Co-efficient of skewness} = \frac{3(\text{Mean} - \text{Median})}{S}$$

Its value lies between -3 and 3. Its value is zero for symmetrical distribution. It is positive for a positively skewed distribution, and negative for a negatively skewed distribution.

Bowley's Co-efficient of Skewness:

Another measure of skewness was suggested by Arthur Lyon Bowley (1869-1957). So it is called Bowley's co-efficient of skewness. It is defined as:

$$\text{Co-efficient of skewness} = \frac{(Q_3 - \text{Median}) - (\text{Median} - Q_1)}{(Q_3 - \text{Median}) + (\text{Median} - Q_1)}$$

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OR

$$\text{Co-efficient of skewness} = \frac{Q_1 + Q_3 - 2(\text{Median})}{Q_3 - Q_1}$$

Its value lies between -1 and 1. Its value is zero for symmetrical distribution, negative for negatively skewed distribution and positive for positively skewed distribution.

Another measure of skewness is defined as:

$$\text{Co-efficient of skewness} = \frac{\mu_3}{\sigma^3} \quad (\text{for population data})$$

$$\text{Co-efficient of skewness} = \frac{m_3}{S^3} \quad (\text{for sample data})$$

The value of this co-efficient lies between -2 & 2 usually. Its value is zero for symmetrical distribution, negative for negatively skewed distribution and positive for positively skewed distribution. Sometimes this coefficient of skewness is denoted by $\sqrt{\beta_1}$.

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Kurtosis:

Karl Pearson introduced the term kurtosis for the degree of peakedness or flatness of a unimodal frequency curve. When the values of a variable are closely bunched around the mode in such a way that the peak of the curve becomes relatively high, we say that the curve is **Lepto-kurtic**. If, on the other hand, the curve is flat topped, we say that the curve is **platy-kurtic**. So the curve which is neither very peaked nor very flat topped is called **Meso-kurtic** and used for comparison.

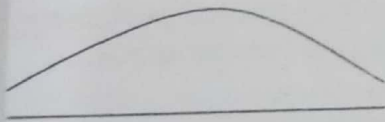
Kurtosis is usually measured by second moment ratio:

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2} \quad (\text{For population data})$$

$$b_2 = \frac{m_4}{(m_2)^2} \quad (\text{For sample data})$$

- If β_2 or $b_2 = 3$ Then curve is meso-kurtic.
 If β_2 or $b_2 < 3$ Then curve is platy-kurtic.
 If β_2 or $b_2 > 3$ Then curve is lepto-kurtic.

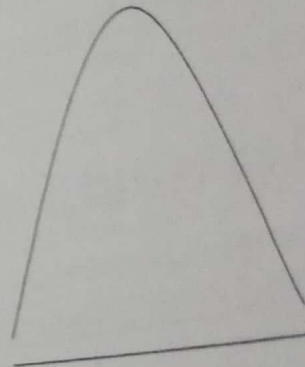
SAIF-UR-REHMAN
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Platy-kurtic
 If β_2 or $b_2 < 3$



Meso-kurtic
 If β_2 or $b_2 = 3$



Lepto-kurtic
 If β_2 or $b_2 > 3$

Percentile co-efficient of kurtosis:

It is defined as:

$$K = \frac{Q.D.}{P_{90} - P_{10}}$$

Its value lies between 0 & 0.50. It is 0.263 for meso-kurtic curve (normal distribution).

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Note: The normal distribution is always symmetrical and meso-kurtic, so for normal distribution: β_1 or $b_1 = 0$ and β_2 or $b_2 = 3$

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STATISTICS HSSC-I

SECTION - A (Marks 17)

Time allowed: 25 Minutes

NOTE - Section-A is compulsory and comprises pages 1-2. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Q. 1 Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) The data which have NOT undergone any statistical treatment are _____
- A Primary data B Secondary data
C Discrete data D Qualitative data
- (ii) An Ogive is a _____
- A Frequency curve B Frequency polygon
C Cumulative frequency polygon D Frequency histogram
- (iii) The process of arranging data into rows and columns is called _____
- A Frequency distribution B Classification
C Tabulation D Array
- (iv) Histogram is a graph of _____
- A Frequency distribution B Time series
C Qualitative data D Ogive
- (v) When a distribution is symmetrical and has one mode, the highest point on the curve is called the _____
- A Mode B Median
C Mean D All of these
- (vi) Sum of squares of deviations of the values is least when deviation is taken from _____
- A Median B Mode
C Mean D Harmonic mean
- (vii) The geometric mean of 'a' and 'b' is _____
- A ab B $(a+b)^2$
C $\sqrt{ab}$ D $\sqrt{a+b}$
- (viii) Mode of the series 2, 2, 2, 3, 3, 3, 2, 3, 3, 4 is _____
- A 3 B 2 and 3
C 4 D None of these

30
If any value in a series is zero, then we can not calculate the _____

- A Mean
B Geometric mean
C Mode
D Median

The positive square root of the variance of a distribution is known as _____

- A Standard deviation
B Mean deviation
C Absolute deviation
D None of these

If $\bar{x} = 5$, then which of the following expressions is minimum?

- A $\sum (x - 25)^2$
B $\sum (x - 5)^2$
C $\sum |x - 5|$
D $\sum |x - 25|$

Standard deviation of 2, 2, 2, 2 and 2 is equal to _____

- A 2
B 8
C Zero
D 4

If the moment Ratio $\beta_2 = 3$ then the distribution is _____

- A Platykurtic
B Positively skewed
C Symmetrical
D Mesokurtic

The price used in the construction of consumer price index numbers is _____

- A The retail price
B The wholesale price
C The fix price
D None of these

Base year weighted index numbers are also known as _____

- A Laspeyre's
B Paasche's
C Fisher's
D None of these

Long term variation is regarded as _____

- A Secular trend
B Seasonal variation
C Cyclical variation
D Irregular variation

The graph of a time series is called a _____

- A Histogram
B Historigram
C Trend line
D Scatter diagram

Examiner's use only:

Total Marks:

17

Marks Obtained:



Roll No.

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Sig. of Candidate.

Answer Sheet No.

35

Sig. of Invigilator.

STATISTICS HSSC-I**SECTION - A (Marks 17)**

Time allowed: 25 Minutes

NOTE: Section-A is compulsory and comprises pages 1 - 2. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q. 1 Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) The data which have already been collected by some one are called _____
- A. Secondary data ☒ B. Primary data
- C. Arrayed data D. None of these
- (ii) A graph of cumulative frequency distribution is called _____
- A. Histogram B. Frequency polygon
- ☒ C. Ogive D. None of these
- (iii) Total angle of Pie chart is _____
- A. 270° B. 300°
- C. 320° ☒ D. 360°
- (iv) If a curve has a longer tail to the left, it is called a _____
- A. Symmetrical curve B. Positively skewed curve
- ☒ C. Negatively skewed curve D. None of these
- (v) If the mean is less than mode, the distribution is _____
- A. Positively skewed ☒ B. Negatively skewed
- C. Symmetrical D. None of these
- (vi) The most central value of an arrayed data is _____
- A. Mode ☒ B. Median
- C. Mean D. Harmonic mean
- (vii) The mode of 2, 10 and 7 is _____
- A. 10 B. 2
- C. 7 ☒ D. None of these
- (viii) In a symmetrical distribution, the mean, median and mode are _____
- ☒ A. Always equal B. Sometimes equal
- C. Never equal D. None of these
- (ix) For a certain distribution if $\sum(x-14) = 5$, $\sum(x-25) = 0$ and $\sum(x-20) = 18$ the value of mean is _____
- ☒ A. Zero B. 25
- C. 18 D. 14

- (x) The variance of a constant is equal to _____
A. Constant
B. Zero
C. Unity
D. None of these
- (xi) If x and y are independent, $Var(x - y)$ is equal to _____
A. $Var(x) + Var(y)$
B. $Var(x) - Var(y)$
C. $Var(x + y)$
D. Zero
- (xii) If the third moment about mean is zero ($m_3 = 0$) the distribution is _____
A. Mesokurtic
B. Leptokurtic
C. Symmetrical
D. Platykurtic
- (xiii) Second moment about mean is always equal to _____
A. 0
B. 1
C. Variance
D. Standard deviation
- (xiv) The index given by $\frac{\sum P_n q_n}{\sum P_o q_o} \times 100$ is _____
A. The Laspeyre's index
B. The Paasche's index
C. The Value index
D. None of these
- (xv) If Laspeyre's index number is 200, Paasche's index number is 200 the Fisher's index number is _____
A. 100
B. 200
C. Zero
D. 400
- (xvi) Increased demand for coolers in summer and heaters in winter is an example of _____
A. Secular trend
B. Cyclical variation
C. Seasonal variation
D. Irregular variation
- (xvii) In chain base method, the base period is _____
A. Fixed
B. Changed
C. Constant
D. None of these

For Examiner's use only:

Total Marks:

17

Marks Obtained:



Roll No.

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Sig. of Candidate. _____

Answer Sheet No. _____

Sig. of Invigilator. _____

STATISTICS HSSC-I**SECTION - A (Marks 17)****Time allowed: 25 Minutes**

NOTE: Section-A is compulsory and comprises pages 1-2. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Q. 1 Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) A specific characteristic of a sample is called:
- | | |
|--------------|---|
| A. Variable | B. Constant |
| C. Parameter | ☒ D. Statistic |
- (ii) A variable that assumes only some selected values in a range is called:
- | | |
|-------------------------|---|
| A. Continuous variable | B. Quantitative variable |
| C. Qualitative variable | ☒ D. Discrete data |
- (iii) Colors of flowers is an example of:
- | | |
|--------------------------|--|
| A. Quantitative variable | B. Symmetric variable |
| C. Skewed variable | ☒ D. Qualitative variable |
- (iv) The grouped data are called:
- | | |
|--|----------------------|
| A. Primary data | B. Raw data |
| ☒ C. Secondary data | D. Difficult to tell |
- (v) In classification, the data are arranged according to:
- | | |
|--|----------------|
| A. Percentages | B. Differences |
| ☒ C. Similarities | D. Ratios |
- (vi) A sector diagram is also called:
- | | |
|---|----------------|
| A. Bar diagram | B. Histogram |
| ☒ C. Pie diagram | D. Historigram |
- (vii) The measure of central tendency listed below is:
- | | |
|------------------|--|
| A. The raw score | ☒ B. The Mean |
| C. The range | D. Standard deviation |
- (viii) Mode of the series 0, 0, 0, 2, 2, 3, 3, 8, 10 is:
- | | |
|------|---------------------------------------|
| A. 2 | ☒ B. 0 |
| C. 3 | D. No Mode |

- (ix) The population mean μ is called:
- A. Discrete variable
 C. Continuous variable
 B. Parameter
 D. Sampling unit
- (x) In Symmetrical distribution $Q_3 - Q_1 = 20$, Median = 15, Q_1 is equal to:
- A. 5
 C. 20
 B. 25
 D. 15
- (xi) The standard deviation of -5, -5, -5, -5 is:
- A. 0
 C. -5
 B. +5
 D. -25
- (xii) The measures of dispersion can never be:
- A. Negative
 C. Positive
 B. Zero
 D. Equal to 2
- (xiii) When index number is calculated for Several variables it is called:
- A. Composite index
 C. Volume index
 B. Wholesale price index
 D. Simple index
- (xiv) Paasche's price index number is also called:
- A. Base year weighted
 C. Simple aggregative index
 B. Current year weighted
 D. Consumer price index
- (xv) Index number having upward bias is:
- A. Laspeyre's index
 C. Fisher's index
 B. Paasche's index
 D. Marshall Edgeworth index
- (xvi) If regression line of $\hat{y} = 5$ then value of regression coefficient of y and x is:
- A. 0.5
 C. 1
 B. 0
 D. 5
- (xvii) A business cycle has:
- A. One stage
 C. Three stages
 B. Four stages
 D. Two stages

For Examiner's use only:

Total Marks:

17

Marks Obtained:

—1HA 1513—



Roll No.

Sig. of Candidate. _____

Answer Sheet No. _____

Sig. of invigilator. _____

STATISTICS HSSC-I
SECTION - A (Marks 17)

Time allowed: 25 Minutes

NOTE: Section-A is compulsory and comprises pages 1-2. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Q. 1 Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) A specific characteristic of a population is called
A. Statistic ☒ B. Parameter
C. Variable D. Sample
- (ii) Listing of the data in order of Numerical Magnitude is called:
A. Raw data ☒ B. Arrayed data
C. Discrete data D. Continuous data
- (iii) The word ogive is also used for:
A. Frequency polygon ☒ B. Cumulative frequency polygon
C. Frequency curve D. Histogram
- (iv) A variable that assumes any value within a range is called:
A. Discrete variable ☒ B. Continuous variable
C. Independent variable D. Dependent variable
- (v) The average of lower and upper class limits is called:
A. Class boundary B. Class frequency
☒ C. Class mark D. Class limit
- (vi) A pie diagram is represented by a:
A. Rectangle B. Triangle
☒ C. Circle D. Square
- (vii) Step deviation method or coding method is used for computation of the:
A. Geometric mean B. Harmonic mean
☒ C. Arithmetic mean D. Weighted mean
- (viii) A curve that tails off to the right end is called:
☒ A. Symmetrical B. Negatively skewed
☒ C. Positively skewed D. Both A and B

- (ix) The sample mean $\bar{X}$ is a:
- A. Parameter
B. Constant
C. Variable
D. Statistic
- (x) The variance is zero only if all observations are:
- A. Different
B. Square
C. Square root
D. Same
- (xi) The range of the values $-5, -8, -10, 0, 6, 10$ is:
- A. 0
B. 10
C. -10
D. 20
- Handwritten: $10 - (-10) = 20$
- (xii) Bowley's coefficient of Skewness lies between:
- A. 0 and 1
B. -2 and +2
C. -1 and 0
D. -1 and +1
- (xiii) Index for base period is always taken as:
- A. 100
B. One
C. 200
D. Zero
- (xiv) Price relative computed by chain Base method is called:
- A. Link relative
B. Chain indices
C. Price relatives
D. None of these
- (xv) Base year quantities as weights are used in:
- A. Laspeyre's method
B. Paasche's method
C. Fisher's ideal method
D. None of these
- (xvi) In simple regression equation, the number of variables involved is:
- A. 2
B. 1
C. 0
D. 3
- (xvii) Depression in business is:
- A. Cyclical
B. Secular trend
C. Seasonal
D. Irregular

For Examiner's use only:

Total Marks:

17

Marks Obtained:

1HA 1613



Roll No.

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Sig. of Candidate.

Answer Sheet No.

Sig. of Invigilator.

47

STATISTICS HSSC-I

SECTION - A (Marks 17)

Time allowed: 25 Minutes

NOTE: Section-A is compulsory. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q. 1 Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) A statistics which is not measureable is called:
A a constant B an attribute C a variable D a parameter
- (ii) The process of arranging data into rows and columns is called:
A frequency distribution B classification
C tabulation D array
- (iii) Graph of a time series is called:
A Histogram B Ogive C Histogram D Polygon
- (iv) If a curve has longer tail to right it is called:
A Positively skewed curve B Negatively skewed curve
C J-shaped curve D Symmetric curve
- (v) When all the values in a series occur the same number of time, then it is not possible to compute:
A Mean B Median C Mode D Weighted mean
- (vi) If $\bar{X} = 100$ and $Y = 2X - 200$ then the Mean of Y values will be:
A 0 B 2 C 100 D 200
- (vii) Sum of squares of deviation of the values is least when deviation are taken from:
A Median B Mode C Mean D Harmonic Mean
- (viii) For a positively skewed distribution:
A Mean > Mode B Mode > Mean C Median > Mean D None of these
- (ix) The Geometric Mean of numbers 2, 4, 8 and 64 is:
A 2 B 4 C 8 D 64
- (x) If $S.D.(X) = 5$, then $S.D.(\frac{2X+5}{2})$ is equal to:
A 5 B 10 C 15 D 15.5
- (xi) For a Moderately skewed distribution which of the following hold:
A $M.D. = \frac{1}{5} S.D$ B $Q.D. = \frac{2}{3} S.D$ C $Q.D. = \frac{5}{6} M.D$ D All of these
- (xii) The distribution is symmetrical if the moment co-efficient of skewness $\sqrt{b_1}$ is:
A Negative B Positive C 0 D 3
- (xiii) If X and Y are independent then $S.D.(X - Y)$ is equal to:
A $\sqrt{\text{var}(X) - \text{var}(Y)}$ B $\sqrt{\text{var}(X) + \text{var}(Y)}$
C $S.D.(X) + S.D.(Y)$ D $S.D.(X) - S.D.(Y)$
- (xiv) If all values are considered of equal importance in calculation of an index number, the index number is:
A Weighted B Simple C Un-weighted D None of these
- (xv) Base year weighted index numbers are given by:
A Laspeyres B Paasche's C Fisher's D C.P.I
- (xvi) If all the points (X, Y) in the scatter diagram lie near a line the correlation co-efficient is:
A Linear B Inverse C Non-Linear D None of these
- (xvii) Long term variations are regarded as:
A Secular trend B Seasoned variations
C Cyclical variations D Irregular variations

For Examiner's use only:

Total Marks:

17

Marks Obtained:

— 1HS 1813 —

Roll No.

Sig. of Candidate.

Answer Sheet No.

Sig. of Invigilator.

50

STATISTICS HSSC-I SECTION - A (Marks 17)

Time allowed: 25 Minutes

Section-A is compulsory and comprises pages 1 - 2. All parts of this section are to be answered on the question paper itself. It should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Circle the correct option i.e. A / B / C / D. Each part carries one mark.

- (i) Census collects the:
☒ A. Primary data ☐ B. Secondary data ☐ C. Fictitious data ☐ D. Official data
- (ii) The word "Statistics" has been derived from the German word:
☐ A. Statista ☐ B. Status ☒ C. Statistik ☐ D. Statistique
- (iii) A variable which can take any possible value in an interval is called:
☐ A. Discrete variable ☒ B. Continuous variable
☐ C. Qualitative variable ☐ D. Finite variable
- (iv) The graph of Time series data is called:
☒ A. Histogram ☐ B. Pie chart ☐ C. Histogram ☐ D. Ogive
- (v) Total of relative frequencies is always:
☐ A. Half ☒ B. One ☐ C. 100 ☐ D. Quarter
- (vi) The sum of deviations from mean is always:
☐ A. Least ☐ B. Maximum ☐ C. One ☒ D. Zero
- (vii) Which of the following averages is not affected by extreme values?
☐ A. Arithmetic mean ☒ B. Median ☐ C. G.M
- (viii) If mean of 5 values is 10, then the sum of the values will be:
☐ A. 2 ☐ B. 15 ☐ C. 25 ☒ D. 50
- (ix) The variance of the values 7, 7, 7, 7, 7 is:
☐ A. 42 ☐ B. 7 ☒ C. Zero ☐ D. $\sqrt{7}$
- (x) If X and Y are two independent random variables, $\text{Var}(x)=4$ and $\text{Var}(y)=9$, then $\text{Var}(2x+y)$ is:
☐ A. 13 ☐ B. 17 ☒ C. 25 ☐ D. 26
- (xi) In a symmetrical distribution, if $Q_1 = 6$ and $Q_3 = 18$ then median is:
☒ A. 12 ☐ B. 15 ☐ C. 24 ☐ D. Zero
- (xii) The empirical relationship between mean, median and mode is: Mode =
☒ A. 3 Mean - 2 Median ☐ B. 2 Mean - 3 Median
☐ C. 3 Median - 2 Mean ☐ D. 2 Median - 3 Mean
- (xiii) The Link relatives are the percentage ratios of current year price and:
☐ A. Previous year quantity ☐ B. Base year quantity
☐ C. Next year price ☒ D. Preceding year price
- (xiv) Which index number helps the Government to formulate economic policies and determine the wages of employees?
☐ A. Whole sale price index ☒ B. Consumer price index
☐ C. Quantity index ☐ D. Simple index
- (xv) The dependent variable is also called:
☐ A. Regressor ☒ B. Explanatory variable
☐ C. Predictor ☐ D. Response variable
- (xvi) The value of correlation coefficient (r) lies between:
☐ A. 0 and 1 ☐ B. -1 and 0 ☒ C. -1 and 1 ☐ D. -2 and 2
- (xvii) Increased demand of soft drinks in summer and woolen clothes in winter season is:
☒ A. Seasonal variation ☐ B. Secular variation
☐ C. Cyclical variation ☐ D. Random variation

$$\begin{aligned} \Sigma x &= n\bar{x} \\ 5 \times 10 &= 50 \\ &= \text{Var}(2x) + \text{Var}(y) \\ 4\text{Var}(x) + \text{Var}(y) \\ 4 \cdot 4 + 9 &= 25 \end{aligned}$$

For Examiner's use only:

Total Marks:

17

Marks Obtained:



STATISTICS HSSC-I
SECTION - A (Marks 17)

53

Time allowed: 25 Minutes

Version Number 1 8 5 5

Note: Section - A is compulsory. All parts of this section are to be answered on the separately provided OMR Answer Sheet which should be completed in the first 25 minutes and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Q. 1 Choose the correct answer A / B / C / D by filling the relevant bubble for each question on the OMR Answer Sheet according to the instructions given there. Each part carries one mark.

- 1) A specific characteristic of a sample is called:
A. Variable B. Constant C. Parameter ☒ D. Statistic
- 2) Census collects the:
A. Official data B. Fictitious data ☒ C. Primary data D. Secondary data
- 3) The number of tally sheet count for each value or a group is called:
A. Class limit ☒ B. Frequency C. Class boundary D. Class width
- 4) Total angle of pie chart is:
A. 270° B. 300° C. 320° ☒ D. 360°
- 5) In a moderately skewed distribution, the mean is 11 and the median is 13 then the value of mode is:
☒ A. 17 B. 13 C. 11 D. 15
- 6) Ten families have an average of 2 boys. How many boys do they have together?
A. 2 B. 10 C. 12 ☒ D. 20
- 7) The geometric mean of the two numbers X_1 and X_2 is 9. If $X_1 = 3$, then X_2 is equal to:
A. 3 B. 9 ☒ C. 27 D. 81
- 8) Mean of 200 times of 2 is:
A. 200 B. 0.01 C. 100 ☒ D. 2
- 9) Population variance is also called:
☒ A. Sigma squared B. Negative sigma C. Square root D. Cubic root
- 10) Standard deviation is always calculated from:
☒ A. Mean B. Median C. Mode D. Lower quartile
- 11) If the price of a kg of mutton was Rs.200/- in 2001 and Rs.350 in 2005, the simple price relative in 2005 is:
☒ A. 175 B. 57 C. 100 D. 350
- 12) The price relative are the percentage ratios of current year price and:
A. Base year quantity B. Previous year quantity
☒ C. Base year price D. Current year quantity
- 13) The dependent variable is also called:
A. Regressor B. Explanatory variable
C. Predictor ☒ D. Response variable
- 14) Regression coefficient and slope of a line are:
A. Dependent ☒ B. Same C. Not same D. Independent
- 15) Which of the following could NOT be the value of r ?
A. -0.683 B. 0.197 ☒ C. -1.369 D. 1.000
- 16) A business cycle has:
A. One phase B. Two phases C. Three phases ☒ D. Four phases
- 17) Value of b in the trend line $Y = a + bX$ is:
A. Always negative B. Always positive
C. Always zero ☒ D. Both negative and positive

(2018)

STATISTICS HSSC-I

SECTION - A (Marks 17)

56

Time: 25 Minutes

Version Number 3 1 3 1

Section - A is compulsory. All parts of this section are to be answered on the separately provided Answer Sheet which should be completed in the first 25 minutes and handed over to the Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Mark the correct answer A / B / C / D by filling the relevant bubble for each question on the OMR Answer Sheet according to the instructions given there. Each part carries one mark.

A specific characteristic of a population is called:

- A. Statistic ☒ B. Parameter C. Variable D. Sample

In a Census _____ is collected.

- A. Official data ☒ B. Fictitious data C. Primary data D. Secondary data

For graphic presentation of a frequency distribution, the paper to be used is:

- A. Butter paper ☒ B. Graph paper C. Ordinary paper D. Carbon paper

A pie diagram is represented by a

- A. Rectangle ☒ B. Triangle C. Circle D. Square

In a moderately skewed distribution, the mean is 11 and the median is 13 then the value of mode is:

- A. 15 ☒ B. 13 C. 11 D. 17

If the arithmetic mean of the two numbers X_1 and X_2 is 5 if $X_1 = 3$, then X_2 is:

- A. 3 ☒ B. 5 C. 7 D. 10

Median from the values 6, 7, 4, 5, 3, 7, 4 is:

- A. 3 ☒ B. 4 C. 5 D. 7

If the distribution has two modes then it is called:

- A. Uni-modal ☒ B. Bi-modal C. Tri-modal D. By-modal

The lack of uniformity or symmetry is called:

- A. Skewness ☒ B. Dispersion C. Kurtosis D. Standard deviation

The range of the scores 29, 3, 143, 27, 99 is:

- A. 140 ☒ B. 143 C. 146 D. 70

If Laspeyre's index number is 200, Paasche's index number is 200, the Fisher's index number is:

- A. 100 ☒ B. 200 C. 20000 D. 400

The Link relatives are the percentage ratios of current year price and:

- A. Previous year quantity ☒ B. Base year quantity
C. next year price D. Preceding year price

The independent variable is also known as:

- A. Regressand ☒ B. Predicted C. Explained D. Predictor

In a simple regression equation, the number of variables involved is:

- A. 2 ☒ B. 1 C. 0 D. 3

When b_{xy} is positive, then b_{yx} will be:

- A. Negative ☒ B. Zero C. Positive D. One

An era of prosperity is an example of:

- A. Secular trend ☒ B. Seasonal variation
C. Cyclical variation D. Irregular variation

The graph of a time series is called a:

- A. Histogram ☒ B. Historigram C. Trend line D. Scatter diagram

STATISTICS HSSC-I SECTION - A (Marks 17)

59

25 Minutes

Version Number 3 1 3 1

Section - A is compulsory. All parts of this section are to be answered on the separately provided Answer Sheet which should be completed in the first 25 minutes and handed over to the Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

Mark the correct answer A / B / C / D by filling the relevant bubble for each question on the OMR Answer Sheet according to the instructions given there. Each part carries one mark.

Graph of a cumulative frequency distribution is called:

- A. Histogram
B. Ogive
C. Frequency curve
D. Frequency polygon

Number of family members in different families is an example of:

- A. Discrete variable
B. Dependent variable
C. Continuous variable
D. Qualitative variable

A set of all units of interest in a study is called:

- A. Sample
B. Population
C. Parameter
D. Statistic

In a statistical table row captions are called:

- A. Box head
B. Stub
C. Body
D. Title

If mean is less than mode, the distribution is called:

- A. Positively skewed
B. Symmetrical
C. Negatively skewed
D. Bimodal

Sum of absolute deviations is least, when deviations are taken from:

- A. Mean
B. Median
C. Mode
D. Geometric mean

Median divides data into:

- A. 2 parts
B. 3 parts
C. 4 parts
D. 5 parts

For two independent variables X and Y , the $S.D.(X - Y)$ is equal to:

- A. $\sqrt{Var(X) + Var(Y)}$
B. $\sqrt{Var(X) - Var(Y)}$
C. $S.D(X) + S.D(Y)$
D. $S.D(X) - S.D(Y)$

A disadvantage of range is that it depends upon:

- A. Absolute deviations
B. Squared deviations
C. Two extreme values
D. Two quartiles

Bowley's coefficient of skewness lies between:

- A. 0 and 1
B. -1 and +1
C. -1 and 0
D. -2 and +2

In chain base method, the base period is:

- A. 1st year
B. Last year
C. Fixed
D. Not fixed

Paasche's index numbers are also called:

- A. Simple aggregative
B. Base year weighted
C. Consumer price
D. Current year weighted

Dependent variable is also called:

- A. Explanatory variable
B. Predictor
C. Regressand
D. Regressor

The value of correlation coefficient lies between:

- A. 0 and 1
B. -1 and 0
C. -1 and +1
D. -0.5 and 0.5

In method of least squares, the sum of squares of residuals is:

- A. Minimum
B. Maximum
C. Zero
D. Negative

A straight line is fitted to the time series, when the variations are:

- A. Linear
B. Non linear
C. Upward
D. Downward

Price hike in the month of Ramazan is an example of:

- A. Secular trend
B. Cyclical variations
C. Seasonal variations
D. Irregular variations